Naveen Kumar P

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Education

Sri Sai Ram Engineering College **Chennai, India**
B.Tech – Information Technology *2023 – 2027 CGPA: 8.06*

Gokilambal Matric Hr. Sec. School
Higher Secondary (12th Grade) *2023 93%*

Skills

Languages: Python, Java, JavaScript, SQL
Web: HTML, CSS, React.js, Bootstrap, Node.js
Databases: MongoDB
Tools & Platforms: Git, GitHub, Firebase, VS Code, Postman
Domains: Frontend Development, Backend Fundamentals, REST APIs, System Design Fundamentals, Computer Networking Fundamentals, AI Applications

Experience

Product Engineering Intern **January 2026 – March 2026**
CODEGRAMMER

- Developed structured video-based learning content using HTML, CSS, and JavaScript by converting provided transcripts and audio into interactive lessons.
- Built a WordPress plugin to connect recruiters and students, enabling job posting and application workflows.
- Integrated resume submission flow by leveraging an external resume-generation plugin for streamlined applications.
- Conducted product reviews of the platform, providing regular feedback and suggesting improvements to enhance usability and workflow.

Frontend Developer Intern **June 2025 – July 2025**
CODEHUNTERS

- Built and modified React.js UI components based on project requirements.
- Participated in code reviews and improved layout responsiveness.

Projects

AI-Powered Learning Tutor Platform | *React , Node.js, Firebase, Gemini, Groq*

- Built a full-stack platform that generates **personalized courses, lessons, and quizzes** using LLMs, implementing **lazy generation and caching** to minimize redundant API usage and improve scalability.
- Designed a **modular backend with multi-provider AI integration (Gemini, Groq)** and rule-based orchestration to handle different tasks (content generation, quizzes) with fallback support for reliability.
- Developed a **rule-based adaptive learning system** using user progress, quiz performance, and spaced revision logic to recommend next lessons, identify weak topics, and guide structured learning paths.

NullId – Secure Anonymous Chat System | *React, Firebase, Web Crypto API*

- Developed a real-time chat application with **client-side encrypted messaging** using ECDH key exchange and AES-GCM via the Web Crypto API.
- Designed a **privacy-focused user model** with selective data visibility and real-time communication features powered by Firestore.

Interactive Portfolio with Custom Terminal & WebGL Rendering | *React, Three js*

- Built an **interactive developer portfolio** with a custom terminal interface, enabling command-driven navigation, dynamic UI control, and feature triggering within a React-based application.
- Developed advanced frontend systems including **WebGL-based 3D visualization (React Three Fiber)** and real-time UI overlays, focusing on performance, component architecture, and immersive user interaction.

Publication

Quantum-Inspired Adaptive AI Tutor for Personalised Learning

- Presented at the *1st International Conference on Quantum Innovations for Computing and Knowledge Systems (QUICK'26)* – Organized by Vellore Institute of Technology (VIT), Chennai in collaboration with Deakin University, Australia | March 2026. doi.org/10.1051/epjconf/202636001020
- Proposed an adaptive learning architecture combining learner-state modelling with AI-driven content generation; explored quantum-inspired optimisation strategies for personalised quiz sequencing and instructional flow.